

EUROPEAN EXPERIENCE IN BIODYNAMIC VITICULTURE FOR GEORGIA

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ABSTRACT

Since the signing of the Association Agreement with the European Union in 2014, Georgia has been required to align its agricultural, and particularly its viticulture and winemaking sectors, with EU regulatory frameworks. This includes the implementation of seedling certification, vineyard zoning, product traceability, and quality management systems. Despite significant institutional reforms and the strengthening of geographical indication (PDO) protection, the sector continues to face structural and technological challenges limiting its integration into the European market.

The study examines the dynamics of reforms implemented through cooperation with the EU and international partners, as well as their impact on the competitiveness of Georgian viticulture. The analysis draws on documentary sources, sectoral statistics, and comparative assessment with successful European viticulture models, particularly the terroir zoning system of Alsace (Domaine Zind-Humbrecht).

The findings indicate that Georgia has achieved progress in PDO system implementation, establishment of certified nurseries, and development of institutional frameworks. However, persistent systemic barriers — land fragmentation, low mechanization, limited digital literacy, and weak adoption of innovative agro-technologies — continue to impede effective transformation. Nonetheless, Georgia maintains unique competitive advantages: millennia-old winemaking culture, rich genetic diversity of indigenous grape varieties (Saperavi, Rkatsiteli, Kisi, Ojaleshi), and diverse terroirs across Kakheti, Imereti, Racha, and Kartli regions.

The sustainable advancement of Georgian viticulture requires a comprehensive modernization strategy integrating precision viticulture, digital traceability systems, farmer training and knowledge transfer, as well as cooperative approaches to overcome structural fragmentation. Combining traditional cultural and genetic assets with EU-compatible innovative practices represents the main pathway for transforming Georgia's wine sector from bulk production toward a premium-oriented, competitive, and sustainable industry within the global wine economy.

Keywords: Georgia; Viticulture; Winemaking; EU Association Agreement; Terroir zoning; PDO (Protected Designation of Origin); Biodynamic viticulture; Organic certification; Precision viticulture; Agro-technologies; Land fragmentation; Wine competitiveness;

Indigenous grape varieties (Saperavi, Rkatsiteli, Kisi, Ojaleshi); Global wine market; Sustainable agriculture

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2014 წელს ევროკავშირთან ასოცირების ხელშეკრულების ხელმოწერის შემდეგ, საქართველომ აიღო ვალდებულება, საკუთარი სასოფლო-სამეურნეო და განსაკუთრებით მევენახეობა-მელვინეობის სექტორები დაახლოოს ევროკავშირის რეგულაციურ ჩარჩოებთან. ეს მოიცავს ნერგების სერტიფიცირების, ვენახების ზონირების, პროდუქტის ტრასაბელობისა და ხარისხის მართვის სისტემების დანერგვას. მიუხედავად მნიშვნელოვანი ინსტიტუციური რეფორმებისა და გეოგრაფიული აღნიშვნების (PDO) დაცვის გაძლიერებისა, სექტორი კვლავ აწყდება სტრუქტურულ და ტექნოლოგიურ გამოწვევებს, რომლებიც ზღუდავენ მის ინტეგრაციას ევროპულ ბაზარზე.

შესწავლილ იქნა ევროკავშირთან და საერთაშორისო პარტნიორებთან თანამშრომლობის ფარგლებში განხორციელებული რეფორმების დინამიკა და მათი გავლენა ქართული მევენახეობის კონკურენტუნარიანობაზე. ანალიზი ეფუძნება დოკუმენტურ წყაროებს, სექტორულ სტატისტიკას და შედარებით ანალიზს ევროპული მევენახეობის წარმატებულ მოდელებთან, განსაკუთრებით ელზასის ტერუარის ზონირების სისტემასთან (Domaine Zind-Humbrecht).

კვლევამ აჩვენა, რომ საქართველომ მიაღწია პროგრესს PDO სისტემის დანერგვაში, სერტიფიცირებული სანერგე მეურნეობების შექმნაში და ინსტიტუციური ჩარჩოების გაძლიერებაში. მიუხედავად ამისა, კვლავ გამოკვეთილია ძირითადი სისტემური დაბრკოლებები — მიწის საკუთრების ფრაგმენტაცია, მექანიზაციის დაბალი დონე, სოფლად მცხოვრები მოსახლეობის ციფრული უნარების შეზღუდულობა და ინოვაციური აგროტექნოლოგიების არასაკმარისი დანერგვა. ამასთანავე, საქართველოს გააჩნია უნიკალური კონკურენტული უპირატესობები: მრავალათასწლოვანი

მელვინეობის კულტურა, ადგილობრივი ვაზის ჯიშების გენეტიკური მრავალფეროვნება (საფერავი, რქანითელი, ქისი, ოჯალეში და სხვ.) და განსხვავებული ტერუარები კახეთის, იმერეთის, რაჭისა და ქართლის რეგიონებში.

ქართული მევენახეობის მდგრადი განვითარება მოითხოვს ყოვლისმომცველი მოდერნიზაციის სტრატეგიის განხორციელებას, რომელიც მოიცავს ზუსტ (precision) მევენახეობას, ციფრული ტრასაბელობის სისტემების დანერგვას, ფერმერთა გადამზადებისა და ცოდნის ტრანსფერის მექანიზმებს, აგრეთვე კოოპერაციულ მიდგომებს სტრუქტურული ფრაგმენტაციის დასაძლევად. ტრადიციული კულტურული და ბიოგენეტიკური რესურსების შენემა ევროკავშირის სტანდარტებთან თავსებად ინოვაციურ პრაქტიკებთან წარმოადგენს ძირითად გზას, რომლის მეშვეობითაც საქართველო შეძლებს მელვინეობის ტრანსფორმაციას მასობრივი წარმოებიდან პრემიუმ სეგმენტზე ორიენტირებულ, კონკურენტუნარიან და მდგრად ინდუსტრიად გლობალურ ღვინის ეკონომიკაში.

საკვანძო სიტყვები: საქართველო; მევენახეობა; მელვინეობა; ევროკავშირთან ასოცირების ხელშეკრულება; ტერუარის ზონირება; გეოგრაფიული აღნიშვნა (PDO — Protected Designation of Origin); ბიოდინამიკური მევენახეობა; ორგანული სერტიფიცირება; ზუსტი მევენახეობა; აგროტექნოლოგიები; მიწის ფრაგმენტაცია; ღვინის კონკურენტუნარიანობა; ადგილობრივი ვაზის ჯიშები (საფერავი, რქანითელი, ქისი, ოჯალეში); გლობალური ღვინის ბაზარი; მდგრადი სოფლის მეურნეობა.

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The 8,000-year history of winemaking began precisely on the territory of Georgia. Blessed by a temperate climate conducive to grape growing, Georgia's landscape has become home to more than 525 indigenous grape varieties. The famous Alazani Valley, set against the backdrop of the Caucasus Mountains, is a winemaking haven, with a combination of well-drained, mineral-rich soils and altitude that contribute to the distinctive flavors and aromas characteristic of Georgian wine. Georgia is the only country where the culture of winemaking, which began 8,000 years ago, has not only never been forgotten, but has also developed and strengthened. The most common ritual of Georgian viticulture is the ancient tradition of winemaking in the qvevri. The use of large conical clay vessels buried in the ground allows for natural temperature regulation during fermentation, giving it a unique texture and complexity. The country's strategic location facilitated trade links with Europe, With Asia and the Middle East. The qvevri has been declared a UNESCO Intangible Cultural Heritage Site, and Georgia's ancient winemaking practice continues to have global appeal. Viticulture and winemaking

are leading sectors in Georgia's agricultural sector. In order for Georgian wine to be highly competitive and provide each farmer with a high income, Georgia needs to share European experience and introduce agro-technological innovations. Let's review examples whose experience sharing will take Georgia's ancient heritage to new level.

NOTABLE MODELS OF BIODYNAMIC VITICULTURE

Château Palmer, Pontet-Canet, Climens (Bordeaux, France) - In the Bordeaux region, several chateau estates have adopted biodynamic practices. They integrate biodynamics into prestigious terroirs. Their move adds legitimacy to biodynamics in top-tier wine regions.

Château de Pommard (Burgundy, France) - This estate has tested biodynamic viticulture (starting with small plots) and expressed intent to move toward full biodynamic conversion. A classic Burgundy property experimenting with biodynamics. Useful as a transitional/scale-up model.

Château Feely (France, Saussignac / Dordogne) - Operated under organic and biodynamic principles. Strong in sustainable wine tourism combined with ecological viticulture. Good example of combining small-scale biodynamics with public engagement.

Domaine Zind-Humbrecht (Alsace, France) - Known for wines where soil and site characteristics come through strongly, aided by biodynamic viticulture. A benchmark for high-quality biodynamic wine in cooler climates.

Bonny Doon Vineyard (California, USA) - Originally one of the first to adopt biodynamic practices in the U.S. Emphasized the connection between soil health and wine character. Shows how biodynamics can be integrated into boutique wine philosophy.

Yealands Estate (New Zealand) - Cited among beautiful biodynamic wineries worldwide. Combines energy innovations and high ecological ambition with biodynamic farming. Good case in a maritime / ocean-adjacent climate.

Quivira Vineyards (California, USA) - Also listed among biodynamic wineries in Architectural Digest list. Uses holistic farming, animal integration, and diverse plantings. Example of mixed-farm + biodynamic vineyard.

From studying these examples, some recurring qualities emerge: Scale + integrity: Some examples show biodynamics can work at larger scales (e.g. Emiliana). Strong terroir expression: The wines reflect their site, as biodynamics tends to enhance soil-vine relationships. Integration with other strategies: Many combine biodynamics with wine tourism, branding, diversified income sources. Top-tier estates adopting it: When prestigious wineries convert, it boosts credibility

(e.g. in Bordeaux). Local adaptation: Success depends on tailoring biodynamic techniques to local climate, soil, pests. Commitment over time: It often requires years of soil rebuilding, transition, and patience. Closest Models to Georgian Viticulture Domaine Zind-Humbrecht (Alsace, France). Similarity: Focus on expressing terroir through biodynamic methods in a cool climate with diverse soils, much like Georgia's regions (Kakheti, Imereti, Racha). Why relevant for Georgia: Georgia's unique grape varieties (Saperavi, Rkatsiteli, Kisi, etc.) can shine globally when biodynamics enhances terroir expression.

Château Feely (France, Dordogne)-Similarity: Small-scale, family-run, biodynamic + wine tourism. This is close to Georgia's family wine cellars and growing enotourism. Why relevant: Georgia can follow this model to merge biodynamic farming with wine routes, guesthouses, and agritourism.

Quivira Vineyards (California, USA)- Similarity: Integrates mixed farming, animals, diverse crops with vines. Why relevant: Georgian winemaking is historically linked to polyculture (wine with orchards, vegetables, livestock). Reviving this strengthens authenticity.

Bordeaux estates (Palmer, Pontet-Canet) → relevant only for prestige benchmarking, but Georgia's strength is authenticity, not copying Bordeaux.

Yealands (New Zealand) → high-tech sustainability + maritime climate (too different from Georgian climate). Which Models Georgia Should Share/Adopt

Georgia can present itself not just as the ancient birthplace of wine, but as the modern cradle of biodynamic terroir winemaking. By blending: Ancient qvevri traditions +; Modern biodynamic practices +; Wine tourism & exports: Georgia could create a unique competitive advantage in global wine markets. Domaine Zind-Humbrecht (Alsace, France)-Founded in 1959 by the union of two long winemaking families. Located in Alsace, a cool-climate region with extraordinary geological diversity (granite, limestone, marl, volcanic, clay soils). Today run by Olivier Humbrecht, the first French Master of Wine, who became a pioneer of biodynamics in France. Certified biodynamic by Biodyvin since the late 1990s. Focus on enhancing soil life and microbiome: composting, cover crops, animal manure, and lunar planting cycles. Minimal intervention in the cellar: spontaneous fermentation with native yeasts, long aging on lees, limited sulfites. Wines are often bottled with minimal filtration, allowing terroir character to dominate.

Key Philosophy-Humbrecht views biodynamics not just as a farming technique, but as a philosophy of listening to nature. Terroir expression is the ultimate goal: each grape variety should transparently reflect its vineyard soil, slope, and microclimate. Critics often describe Zind-Humbrecht wines (especially Riesling, Gewürztraminer, Pinot Gris) as "profoundly site-driv-

en". Impact -International recognition: Zind-Humbrecht is often cited as one of the world's greatest biodynamic wine estates. Helped elevate Alsace as a terroir-focused fine-wine region, competing with Burgundy and Mosel. Demonstrates that ancient grape varieties (not just global ones like Chardonnay or Cabernet) can achieve world-class prestige if grown biodynamically and marketed with terroir emphasis.

Why This Model is Relevant for Georgia -Terroir Diversity - Just like Alsace, Georgia has enormous soil and climate diversity: Kakheti's sun-drenched valleys, Imereti's humid hills, Racha's high-altitude sites. Zind-Humbrecht shows how biodynamics can highlight micro-terroirs, turning diversity into a premium brand strength. Indigenous Grapes - Alsace built global prestige on grapes like Riesling, Gewürztraminer, Pinot Gris — not international blockbusters, but regional identities. Similarly, Georgia has over 500 indigenous grape varieties (Saperavi, Rkatsiteli, Kisi, Tsolikouri, Usakhelauri). Biodynamics can elevate these into the same prestige terroir wines category. Minimal Intervention Philosophy - Georgia's qvevri winemaking already aligns with natural and low-intervention practices. Combining biodynamics with qvevri can create a unique global identity: "ancient natural winemaking meets modern biodynamics." Export Competitiveness - Alsace successfully positioned its wines in premium export markets through terroir-driven storytelling. Georgia can do the same: instead of competing on bulk exports, position biodynamic qvevri wines as high-value artisanal products. Climate Adaptation - Alsace, like Georgia, faces challenges of climate change (warmer vintages, unpredictable weather). Zind-Humbrecht shows how biodynamic soil health helps vines adapt and remain resilient, an increasingly urgent lesson for Kakheti's hot summers. Strategic Takeaway for Georgia - Adopting the Zind-Humbrecht model means: Emphasize terroir diversity as a unique selling point; Elevate indigenous grapes as world-class through biodynamics; Blend ancient (qvevri) + modern (biodynamic) for a singular wine identity. Position Georgia as not just the cradle of wine, but the future of terroir-driven biodynamic winemaking.

The diversity of soils allows the estate to match grape variety to site for maximum expression. Each wine is marketed as a reflection of its terroir, making even the same grape variety (e.g. Riesling) taste radically different depending on vineyard. This creates complexity and prestige, since terroir-specific wines are highly valued by collectors and sommeliers. Georgian Vineyards and Soils - Georgia is no less diverse — in fact, it has greater terroir potential than Alsace because of its geography between the Black Sea and Caucasus Mountains. Some highlights: **Kakheti → alluvial soils with clay, sand, and gravel; warm continental climate; ideal for Saperavi (deep, structured reds)**

and Rkatsiteli (rich whites). Imereti → clay-limestone soils, humid climate; produces lighter, fresher wines (Tsitska, Tsolikouri). Racha-Lechkhumi → high-altitude, schist and alluvial soils; famous for Usakheauri, Aleksandrouli — aromatic, elegant reds. Kartli → volcanic and limestone soils, mixed climates; good for sparkling bases and experimental varieties.

Adjara & Guria (Black Sea zones) → sandy-clay soils, maritime influence; potential for fresh whites and rare local grapes.

Prerequisites for terroir expression in Georgia: The soil diversity + native grape diversity (500+ varieties) is unmatched. Georgia has both warm valleys for powerful reds and cool highlands for aromatic whites. Ancient qvevri winemaking already emphasizes authentic terroir identity, much like Alsace's site-specific approach. Can Georgia Be as Successful and Competitive? Yes — Georgia absolutely has the resources, but success depends on how terroir is communicated and marketed - What Georgia has in common with Alsace: Soil diversity → wide spectrum of terroir wines. Indigenous grapes → strong identity (like Riesling in Alsace). Small family wineries → authenticity, uniqueness. History & tradition → qvevri is as prestigious a story as Alsace's terroir.

What Georgia must develop (learning from Zind-Humbrecht): Biodynamic & organic certification → builds credibility in premium export markets. Single-vineyard labeling → show terroir differences (e.g. "Rkatsiteli – Mukuzani slope – limestone soil"). Soil mapping and research → Alsace invested heavily in terroir science; Georgia needs systematic vineyard zoning. Premium positioning → shift away from bulk wine export to terroir-driven fine wines.

Institutional & Certification Barriers - Limited domestic infrastructure: Georgia lacks strong local certification bodies accredited internationally (like Demeter for biodynamics or EU-certified organic inspectors). High cost of foreign certification: Many small family wineries cannot afford to hire international auditors from Germany, Austria, or France. Complex bureaucracy: Export-oriented wineries must comply with EU, US (NOP), and other markets' rules — but Georgia doesn't have unified state support for guiding producers through this.

Knowledge & Awareness Gaps - Low technical expertise: Many Georgian winemakers know natural farming from tradition, but lack formal training in biodynamic principles (soil microbiome, lunar calendars, composting, etc.). Confusion between "natural" and "certified": Georgian qvevri wines are often marketed as "natural," but without certification they can't compete in EU/US premium shelves where consumers look for an organic/biodynamic label. Few specialists: Limited number of agronomists trained in soil health, biodynamics, organic pest management.

Economic & Structural Constraints - Small-scale vineyards: Most Georgian wineries own <5 hectares, making certification costly per hectare. Fragmented land ownership: Thousands of tiny family plots → hard to apply uniform standards. Transition period losses: Organic conversion takes 3 years of chemical-free farming, during which yields may fall while costs rise. Many farmers cannot absorb this financially.

Market & Policy Challenges - Low domestic demand: Georgian consumers are price-sensitive; organic/biodynamic wine sells for higher prices mainly abroad. Export focus on quantity, not quality: Big wineries often prioritize bulk exports to Russia & CIS over premium EU/US organic markets. Weak government incentives: Unlike the EU, Georgia offers little in the way of subsidies, tax relief, or grants for organic/biodynamic conversion. **Environmental & Practical Obstacles** - Disease pressure: Georgia's humid regions (Imereti, Guria, Adjara) face high fungal risks (mildew, rot), making chemical-free farming harder. Climate change: Extreme weather (hail, drought, frosts) pushes farmers toward chemical "insurance." Infrastructure gaps: Limited access to organic preparations, compost, biodynamic sprays (many need to be imported).

What Hinders Development? Lack of affordable, local certification services; Weak state support & subsidies; Economic vulnerability of smallholders; Knowledge gap about global organic/biodynamic standards; Market still driven by bulk exports & price, not terroir-driven premium wines; Strategic Opportunity for Georgia - Establish a Georgian Organic/Biodynamic Certification Body recognized by EU. Provide financial aid for small winemakers during 3-year conversion. Create training centers (with European experts) for farmers on soil health and biodynamic viticulture. Promote indigenous grapes as premium organic wines in EU/US/Japan markets (where demand for organic wine is growing fastest). **What is Single-Vineyard Labeling?** Wines are bottled and marketed from one specific vineyard plot, often with its name, soil type, altitude, or microclimate on the label. Current practice: A few boutique wineries experiment with vineyard-specific wines (e.g., Saperavi from a single Kakhetian slope). Market trend: Most exports are region-based or generic blends, not vineyard-designated wines.

Challenges in Georgia - a) Land Structure & Ownership: Vineyards are highly fragmented after Soviet collectivization. One "vineyard" might be split among many families, making single-vineyard branding legally and practically difficult. Lack of Zoning & Mapping - Georgia lacks a comprehensive terroir classification like Burgundy's climats or Alsace's Grand Crus. Without official vineyard maps, consumers can't trust the difference between one slope and another.

Why It Matters for Georgia - Premium positioning: Vineyard-specific wines could move Georgian exports

from \$3–5 bottles to \$20–50 premium wines. Terroir storytelling: Georgia's diverse soils (alluvial Kakheti, limestone Imereti, volcanic Kartli, alpine Racha) deserve individual recognition. Comparison with France/Italy: Georgia could create its own equivalent of Burgundy's "climats" or Piedmont's "crus," based on ancient terroir traditions and modern research.

Georgia has everything — diverse soils, unique grapes, ancient qvevri tradition — to become as terroir-driven as Burgundy or Alsace. But challenges in land fragmentation, lack of zoning, weak regulation, and limited marketing vision slow progress. If solved, single-vineyard labeling could be a game-changer for Georgia's premium wine competitiveness.

Soil Mapping and Research in Viticulture - What Alsace Did Right - Alsace is an excellent model because it moved from being seen as a "secondary" French region to one of the world's most terroir-respected wine areas. How? Systematic geological surveys (1970s–1990s): Alsace commissioned detailed soil maps, identifying 13 distinct geological formations (granite, schist, limestone, marl, volcanic, etc.).

Zoning vineyards into Crus: Based on soil + micro-climate + slope, Alsace officially classified 51 Grand Cru sites, each with defined boundaries.

Scientific + cultural integration: These maps were not just for academics; they were used by winemakers, regulators, and marketers. A Riesling from "Rangen de Thann" (volcanic soil) could be priced much higher because buyers trusted the terroir designation.

Result: Alsace wines became terroir-driven premium wines. Instead of generic "Alsace Riesling," consumers began to ask for specific terroirs — like Burgundy lovers ask for "Meursault" or "Chambertin."

Why Soil Mapping Matters for Georgia - Premium positioning: Instead of selling Saperavi as a generic red, Georgian wineries could market "Saperavi – Mukuzani, limestone slope, 450m altitude." This immediately signals terroir prestige. Scientific credibility: Buyers and critics (Decanter, Wine Spectator) trust wines that have recognized terroir research. Climate adaptation: Mapping soils and microclimates helps farmers choose the right grape for the right site in the face of climate change (e.g., planting white grapes in higher, cooler zones). Economic benefit: Single-vineyard and terroir-specific wines can sell for 3–10x higher prices than generic PDO wines.

Challenges for Georgia - Fragmented vineyard ownership → difficult to standardize zoning.

High cost of mapping → requires government, universities, and donors.

Limited expertise → Georgia has strong tradition, but needs modern GIS, geology, and soil science.

Political will → PDO expansion is bureaucratic; it may face resistance from large producers who prefer flexibility.

Roadmap for Georgia - Pilot soil mapping project: Select 3 iconic regions (e.g., Kakheti – Mukuzani, Racha – Usakhelauri, Imereti – Tsolikouri). Map soils in detail with international experts + local universities. Create "micro-zones": Introduce sub-PDOs (like "Mukuzani – Tsarapi slope") that are legally protected. National Soil & Climate Database: Build a public platform (similar to France's INAO maps) accessible to winemakers, buyers, and tourists.

International recognition: Partner with OIV (International Organisation of Vine and Wine) to validate Georgia's terroir research. Link with biodynamics: Use zoning to show where biodynamics enhances terroir expression most (e.g., limestone soils in Imereti).

Georgia has all the raw material: diverse soils, ancient grape varieties, rich history. What's missing is systematic soil mapping and terroir zoning. Alsace shows that when a region invests in terroir science, it can shift from being "regional" to being seen as world-class. If Georgia follows this path, its wines could command premium prices and global prestige — not only as "the cradle of wine," but as a 21st-century terroir leader.

Georgian Terroir Atlas – Proposed Structure - Georgia as the cradle of wine: 8,000 years of tradition, 500+ native grapes. Methodology: Soil sampling, GIS mapping, climate modeling, historical vineyard analysis. Definition of terroir: interaction of soil, climate, topography, grape variety, and human practice. Pedology: soils and their structure (sand, clay, limestone, volcanic, schist). Climatology: rainfall, temperature ranges, winds, altitude. Viticultural impact: how soil + climate determine acidity, tannins, alcohol, aromatic profile. Comparison: Burgundy / Alsace vs. Georgia → lessons for positioning. Georgian Wine Regions (Macro-Zones)- Each region gets a full chapter with maps, soils, grapes, and style:

Kakheti (East Georgia) - Main soils: alluvial, loam, gravel, limestone patches. Climate: warm continental, long summers, irrigation challenges. Grapes: Saperavi, Rkatsiteli, Kisi, Khikhvi. Key terroirs: Mukuzani, Kindzmarauli, Tsinandali, Napareuli. Micro-zoning potential: limestone slopes of Mukuzani → powerful Saperavi.

Kartli (Central Georgia) - Soils: volcanic, clay-limestone. Climate: moderate, some continental extremes. Grapes: Chinuri, Tavkveri. Potential: Sparkling wine terroirs (similar to Champagne altitude + limestone).

Imereti (West-Central) - Soils: limestone-clay, fertile valleys. Climate: humid subtropical.

Grapes: Tsolikouri, Krakhuna, Tsitska. Terroir focus: crisp, mineral whites (like Loire or Alsace).

Racha-Lechkhumi (Northwest Highlands) - Soils: schist, slate, alluvial. Climate: cooler, high-altitude terroir. Grapes: Usakhelauri, Aleksandrouli, Mujuretuli. Terroir: elegant, aromatic reds → "Georgia's Piedmont."

Adjara, Guria, Samegrelo (Black Sea regions)

- Soils: sandy-clay, coastal alluvium. Climate: maritime-humid, high rainfall. Grapes: Chkhaveri, Ojaleshi. Terroir: potential for fresh, ocean-influenced wines (like Vinho Verde or Galicia).

Soil Maps & Classification - High-resolution GIS maps showing soil layers. Overlay with vineyard locations. Soil → wine style matrix (e.g., limestone = high acidity, granite = aromatic intensity). Saperavi on clay vs. limestone. Rkatsiteli on alluvial vs. volcanic soils.

Climate Analysis - Heat summation maps (Growing Degree Days). Rainfall distribution.

Frost and hail zones. Climate change predictions (2030–2050). Grape × Terroir Pairings; For each indigenous grape, list best soil & climate matches. Example: Saperavi → thrives on limestone + clay, full-bodied, age-worthy reds. Rkatsiteli → prefers alluvial/clay, gives structured whites, great for qvevri. Tsolikouri → limestone soils, elegant mineral whites. Usakhelauri → schist + altitude, perfumed reds.

Micro-Zoning & Single-Vineyard Potential - Proposal for new PDO sub-categories (like Alsace Grands Crus or Burgundy climats). Identify 20–30 “Grand Cru of Georgia” vineyards by soil + history + quality. Example: “Tsarapi slope” (Mukuzani PDO) → limestone-rich, great for premium Saperavi. “Okureshi terraces” (Lechkhumi) → schist soil, ideal for Usakhelauri.

Alsace: terroir mapping + Grand Cru success. Burgundy: single-vineyard prestige & price impact. Chile (Emiliana): large-scale organic biodynamics. Australia (Cullen): climate adaptation with biodynamics.

Policy & Industry Recommendations - Create a National Terroir Research Center; Government support for soil mapping + climate studies; Incentives for single-vineyard labeling and PDO micro-zones; Subsidies for organic/biodynamic transition. International partnerships with OIV, EU terroir research institutes.

A Georgian Terroir Atlas would: Scientifically prove Georgia’s terroir diversity. Support creation of single-vineyard wines. Justify premium pricing in global markets. Combine ancient qvevri heritage with modern terroir science → a unique competitive identity.

Key Statistics for Georgian Wine Sector (Last ~10 Years) - Production & Grapes

Over the last decade, Georgia’s average annual grape production has been about 223.6 thousand tons, much of which is processed into wine. In 2019, Georgian wine production was reported as ~14 million decaliters (i.e. 1.4 million hectoliters). In the OIV’s 2023 global wine production report, Georgia’s wine production is estimated at 1.9 million hectoliters, a decline of ~3.0 % vs. 2022. In 2024, preliminary reports claim a 27 % increase year-on-year, with production reaching 236 million liters (i.e. ~2.36 million hl), placing Georgia 16th globally in production. Exports & International Trade. In 2021, Georgia exported 107 million bottles of wine to 62 countries, earning ~USD 250 million in revenue.

Historically, in 2011, Georgia exported about 19 million bottles (~USD 50 million). Jamnews in English

From January to September 2024, Georgia exported 73.6 million liters of wine, valued at USD 214.5 million — a ~9 % increase in volume and ~11 % increase in value over same period in 2023. In 2024, exports (wine & spirits) reached nearly 95 million litres, valued at USD 276.1 million, to 72 countries — representing ~6-7% growth vs 2023. In the first eight months of 2024 alone, over 67,000 tons of wine (USD 195.8 million) were exported — a 16 % increase compared with the same period in 2023.

Share, Growth & Economic Impact - Wine comprised ~21 % of Georgia’s total agricultural export value in 2021. Over the period 2012–2021, wine exports grew on average ~21 % in volume and ~22 % in value annually. In 2023, wine exports represented ~0.85 % of GDP. If spirits are included, the share rises up to ~1.50 %. That compares to ~0.64 % in 2015. In 2020, Russia accounted for ~55 % of the value of Georgian wine exports (USD ~130.7 million) — reflecting heavy dependence on that market. In recent years, diversification has increased: Russia’s share in exports dropped somewhat (to ~56 %) in 2021, the lowest since Georgian wine re-entry to Russia. In 2021, 441 companies exported wine — showing how fragmented and numerous the wine businesses are. The number of Georgian wineries has also exploded: in 2006 there were ~80 registered wineries; by 2018 that number was 961; by 2020 ~1,575.

Georgia’s wine exports have surged in both volume and value over the past decade, even from modest starting levels (e.g. ~19 million bottles in 2011). The growth rates (20+ % per year in many years) are among the strongest in the wine world. The growth in number of wineries and exporting companies suggests that Georgia is broadening its base — not relying solely on a few large producers. Also, its export markets are slowly diversifying beyond Russia, although Russia remains highly significant.

Still a modest share globally - Even with growth, in global scale Georgia is still a mid-tier player. Its production is in the low millions of hectoliters, compared to the major wine-producing countries which produce tens (or hundreds) of millions of hectoliters.

High dependence on export & certain markets - The sector is export-oriented, and has been historically very vulnerable to geopolitical shocks (e.g. the Russian embargoes). That dependence remains a risk. But the recent data suggest shift — with exports to the U.S. rising significantly in 2024 (58 % increase in volume over 2023).

Wine is no longer marginal: it forms a nontrivial share of agricultural export value and contributes to GDP. The rise from 0.64 % of GDP in 2015 to ~0.85 % (or more with spirits) in 2023 underscores increasing

economic relevance.

After signing the Association Agreement with the European Union, Georgia is obliged to take into account all the articles of the agreement, and accordingly implement European reforms in the agricultural sector. A transitional decade has passed, certification of seedlings and the introduction of zoning have come into force. Georgia has taken a number of steps in this regard, achieving some success in the world market. There are many obstacles for Georgian farmers in this regard, in particular, innovative approaches, competence, lack of introduction of agro-technologies, land fragmentation, lack of agricultural machinery, and difficulty of integration with the digital world for elderly citizens. Given this, Georgian viticulture is not so competitive that the wine made from grapes obtained from it is highly competitive in the European market.

To solve this, it is necessary to implement a comprehensive modernization strategy for Georgian viticulture and winemaking, aligned with EU standards and international best practices. Such a strategy should include the following key dimensions:

Systematic Vineyard Zoning and Soil Mapping - Completion of scientific terroir zoning, identifying micro-regions with unique soil, climate, and grape variety suitability. Establishment of a "Georgian cru system" to strengthen PDO protection and enable single-vineyard labeling. Adoption of EU-compatible geographical indication (GI) frameworks, ensuring transparency and authenticity in the export market.

Seedling Certification and Genetic Preservation - Expansion of certified nurseries for vine seedlings to guarantee disease-free, high-quality planting material. Protection and propagation of indigenous grape varieties (Saperavi, Rkatsiteli, Kisi, Ojaleshi, etc.) through modern biotechnologies and clonal selection. Integration with EU genetic databases to protect Georgia's ampelographic heritage from misuse.

Adoption of Agro-Technologies - Introduction of precision viticulture (drones, sensors, GIS monitoring) for vineyard management. Use of smart irrigation systems, soil sensors, and weather stations to adapt to climate change. Mechanization of vineyard operations, particularly in fragmented plots, through cooperative machinery pools or leasing programs.

Farmer Education and Knowledge Transfer - Establishment of regional viticulture training centers offering continuous education in EU-compatible practices. Digital literacy programs for farmers, especially elderly vineyard owners, enabling them to participate in traceability systems and online certification platforms. Stronger collaboration between universities, research institutes, and small wineries to promote innovation. **Institutional and Market Reforms** - Development of an integrated certification and traceability system, ensuring compliance with EU organic and biodynamic

standards. Support for wine cooperatives to overcome land fragmentation and strengthen bargaining power. Enhanced marketing campaigns focusing on terroir, sustainability, and authenticity of Georgian wines in EU and global markets. **Financial and Policy Support** - Expansion of state subsidies and EU co-funding programs for vineyard renewal, organic conversion, and certification costs. Long-term investment policies to encourage foreign direct investment in sustainable viticulture projects. Risk insurance mechanisms for farmers facing climate hazards (frosts, hail, droughts).

Georgia has made substantial progress since signing the EU Association Agreement, especially in certification and zoning. However, without deep structural reforms—mechanization, digital integration, farmer training, and a terroir-based branding strategy—Georgian viticulture will continue to lag behind European competitors. The country must now transition from fragmented, traditional practices to a globally competitive viticulture model rooted in its unique biodiversity, cultural heritage, and modern agro-technologies.

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